



**College of Computing and Informatics
Department of Computer Science**

Course Syllabus

Course Title Cloud Computing

**Course Code
1501454**

Fall 2025/2026

Course Syllabus

Course Title	Cloud Computing				
Course Code	1501454				
Credit Hours	3	Contact Hours	3		
Course Category	Program required				
Course Type	Compulsory				
Pre-requisite(s)	1501215 - Data Structures OR 1501214 - Programming With Data Structures OR 1410214 - Programming With Data Structures				
Co-requisite(s)					
Semester	Fall 2025/2026				
Instructor Name	Email	Classroom	Section	Session Type	Time
Karam Mohamed Goda Mohamed Sallam	ksallam@sharjah.a c.ae	W8: Sciences (Women)-W8-STU008M	61	Lecture Undergraduate	1100 - 1215
Karam Mohamed Goda Mohamed Sallam	ksallam@sharjah.a c.ae	W9: Engineering (Women)-TH006	62	Lecture Undergraduate	1100 - 1215

Course Description:

This course introduces students to widely used parallel and distributed techniques and applications of cloud computing – including the related state-of-the-art technologies, algorithms and tools. It allows students to develop understanding of the advanced cloud-based software development skills and to combine their existing and new skills in a real-life large-scale distributed business context.

Course Objectives/Goals:

The goals of this Course are to:

- Be familiar with the fundamental concepts of cloud computing, and cloud computing deliver and deployment models.
- Examine concepts related to virtualization, consolidation, and big data.
- Gain hands-on experience with cloud computing technologies by using cloud computing service providers such as Amazon Web Services (AWS).
- Explore cloud-based applications to understand their architecture, deployment strategies, and integration with existing systems.
- Build cloud-based applications that are highly available, scalable, and secure.

Course Learning Outcomes:

By the end of successful completion of this Course, the student will be able to:

1. Understand cloud computing concepts, architectural principles and constraints.
2. Explain various cloud services and deployment models
3. Apply cloud computing architectural best practices.
4. Create cloud services using a cloud management platform such as AWS.

5. Design architectures based on the main pillars of Cloud Computing: security, reliability, performance efficiency and cost optimization.
6. Test the challenges involved in developing large-scale distributed systems and applications with cloud computing.

Alignment of Course Student Learning Outcomes to Program Student Learning Outcomes

Program Learning Outcomes (PLOs)	Course Learning Outcomes (CLOs)
Analyze a complex computing problem and apply principles of computing and other relevant disciplines to identify solutions.	- Understand cloud computing concepts, architectural principles and constraints. - Apply cloud computing architectural best practices. - Create cloud services using a cloud management platform such as AWS.
Design, implement, and evaluate a computing-based solution to meet a given set of computing requirements in the context of the program's discipline.	- Apply cloud computing architectural best practices. - Design architectures based on the main pillars of Cloud Computing: security, reliability, performance efficiency and cost optimization. - Test the challenges involved in developing large-scale distributed systems and applications with cloud computing.
Apply computer science theory and software development fundamentals to produce computing-based solutions.	- Understand cloud computing concepts, architectural principles and constraints. - Explain various cloud services and deployment models - Apply cloud computing architectural best practices. - Create cloud services using a cloud management platform such as AWS. - Design architectures based on the main pillars of Cloud Computing: security, reliability, performance efficiency and cost optimization. - Test the challenges involved in developing large-scale distributed systems and applications with cloud computing.

Weekly Distribution of Course Topics/Contents

Week	Topic	Comments*	CLOs
1	- Introduction to cloud computing and cloud computing architecture - Grid, Cloud and peer-to-peer computing, and their enabling technologies	Practical activities will be demonstrated every week	1
2	- Cloud computing deployment models - Cloud Computing Service models		1,2
3	- Cloud computing services- Storage		1,2,3,6
4	- Cloud Computing services - Compute		1,2,3,6
5	- Cloud Computing services - Database		1,2,3,6

Week	Topic	Comments*	CLOs
6	- Cloud Computing services – Network.		1,2,3,6
7	- Main Pillars of Cloud Computing Architecture: Security, reliability, Performance efficiency and cost optimization.		4,5,6
8	- Cloud Security and Security Mechanisms		4,5,6
9	- Cost Metrics and Pricing Models		4,5,6
10	- System design for high availability		4,5,6
11	- Scaling and event-driven autoscaling		4,5,6
12	- Load balancing strategies that are used for large scale applications.		4,5,6
13	- Building decoupled architecture on the cloud		4,5,6
14	- Infrastructure automation (Infrastructure as a software)		4,5,6
15	- Project Presentation		1,2,3,4,5,6
16	- Final Exam		

Students' Assessment:

Students are assessed as follows:

Assessment Tool(s)**	Date	Weight (%)
Lab	Throughout the semester (8 to 10 practical AWS labs will be demonstrated)	10
Quizzes	Weeks 4, 9 and 12	10
Final Project	Throughout the semester	15
Midterm Exam	Midterm exams period	20
Final Exam	Week 16	45
Total		100

Course Outcome Assessment Plan:

Course LOs	Teaching/Learning Method(s)	Assessment Tool(s)	Performance Indicators	UoS competencies	Other competencies
Understand cloud computing concepts, architectural principles and constraints.	- Lab session - Lectures - Problem-solving	- Exams - Projects - Lab Exercises	70% of the students achieve this outcome.	- Critical and analytical thinking - Problem solving	
Explain various cloud services and deployment models	- Lab session - Lectures - Problem-solving	- Exams - Projects - Lab Exercises	70% of the students achieve this outcome.	- Critical and analytical thinking - Problem solving	

Course LOs	Teaching/Learning Method(s)	Assessment Tool(s)	Performance Indicators	UoS competencies	Other competencies
Apply cloud computing architectural best practices.	- Lab session - Lectures - Problem-solving	- Exams - Projects - Lab Exercises	70% of the students achieve this outcome.	- Digital literacy - Problem solving	
Create cloud services using a cloud management platform such as AWS.	- Lab session - Lectures - Problem-solving	- Exams - Projects - Lab Exercises	70% of the students achieve this outcome.	- Digital literacy - Problem solving	
Design architectures based on the main pillars of Cloud Computing: security, reliability, performance efficiency and cost optimization.	- Lab session - Lectures - Problem-solving	- Exams - Projects - Lab Exercises	70% of the students achieve this outcome.	- Digital literacy - Problem solving	
Test the challenges involved in developing large-scale distributed systems and applications with cloud computing.	- Lab session - Lectures - Problem-solving	- Exams - Projects - Lab Exercises	70% of the students achieve this outcome.	- Digital literacy - Problem solving	

Scope and Nature of Assessment Tools:

Practical Aspect of the course:

The students expected to work on a project. A project includes all phases of analyzing, critically discussing, designing and implementing a legacy system migration into a cloud one. The instructor is to give the students a set of tools (see the Required Teaching Resources section).

Also, the students will complete a number of practical labs from AWS cloud provider.

In-class Activities/Teaching Methodology:

Lecturing, Q&A on the board, group discussion, system design and implementation using required teaching resources, team-based work on problems (peer learning).

Exams:

Two exams (Midterm and Final) will be conducted during the semester. The Midterm exam will take place during the Midterm Exam period and will cover all the material prior to the Midterm. The Final exam will be conducted in Week 16 and will cover all the material included in the course, with a higher weight given to the material covered after the Midterm.

Teaching and Learning Resources:

Textbook(s):

Thomas Erl and Eric Barcelo Monroy. (2023). Cloud Computing: Concepts, Technology, and Architecture. - 2nd ed. Pearson.

Recommended Readings:

Saurabh Shrivastava, Neelanjali Srivastav, Alberto Artasanchez, Imtiaz Sayed. (2023). AWS for Solutions Architects: The definitive guide to AWS Solutions Architecture for migrating to, building, scaling, and succeeding in the cloud. 2nd ed. Packt Publishing.

Hwang, K., Dongarra, J., and Fox, G.. (2011). Distributed and Cloud Computing: From Parallel Processing to the Internet of Things. 1st ed. Morgan Kaufmann.

Ruparelia, N.. (2016). Cloud computing. ed. The MIT Press.

Attendance policy:

Attendance is compulsory. A student missing 10% of the total allocated course hours will receive 1st warning notice and a student missing 15% will receive 2nd warning notice. A student missing 20% will be forced to withdraw (in accordance with the university regulations).

Plagiarism/Cheating:

Students are expected to do their own work. You are allowed to work on assignments in teams only if specified by the instructor. In other words, students are encouraged to communicate about general principles of the course, but all assigned homework must be done on an individual basis. The instructor is available to provide any assistance that you may need. Cheating is considered a serious offense by the university. You should be aware of the severe penalty for cheating (refer to the student code of conduct published in the university catalogue).

Grading rubrics:

Grading Rubric for Course Projects					
Scaled Score:		1/0	2	3	4
Criteria	Weight	Below Standard	Approaching Standard	At Standard	Exceeds Standard
Introduction	10%	No clear introduction, absent, or no evidence.	The introduction states the main topic but does not adequately preview the structure of the report.	The introduction states the main topic and previews the structure of the report.	The introduction is engaging, states the main topic and previews the structure of the report.
Data	20%	Data are inadequate and selected following the wrong procedure.	Data are selected but procedure of data selection is not followed.	Data selected following the standard procedure but not described in report.	Data are correctly selected following standard procedure and adequately described in report.
Analysis and Reporting of Results (if applicable)	20%	Calculated wrong measures and analysis not relevant.	Calculated correct measures but analyses are not correct.	Calculated correct measures but analyses are not adequate.	Calculated correct measures and analyses are correct and provide more insights.

Grading Rubric for Course Projects

Scaled Score:		1/0	2	3	4
Criteria	Weight	Below Standard	Approaching Standard	At Standard	Exceeds Standard
Authenticity and Citation*	20-30%	Some contents are paraphrased; plagiarism is most likely.	Some content is paraphrased; plagiarism is likely.	Some content is paraphrased and no suspicion of plagiarism.	All contents are original and well written.
Writing	20-30%	Errors in language that cause the content to be mostly incomprehensible.	Some errors in language that often inhibit comprehension.	Some errors in language that do not often inhibit comprehension.	Almost no error in language and well written.
Teamwork	10%	Team did not collaborate or communicate well.	Team did not collaborate or communicate well. A lack of mutual respect was frequently noted.	Team did collaborate or communicate well.	Team did collaborate or communicate well with high level mutual respect.